The Role of Mathematics in Machine Learning

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Leads to:** How to learn the maths for ML; Machine learning.

1. Overview

Key point: Four branches of mathematics carry most of ML: **linear algebra** (G-1090) stores the data, calculus improves the model, probability decides when the data is unclear, and statistics draws conclusions from the data.

ML is not magic: every algorithm is mathematics running on data. To understand why an algorithm works, and to fix it when it does not, we need the mathematics underneath.

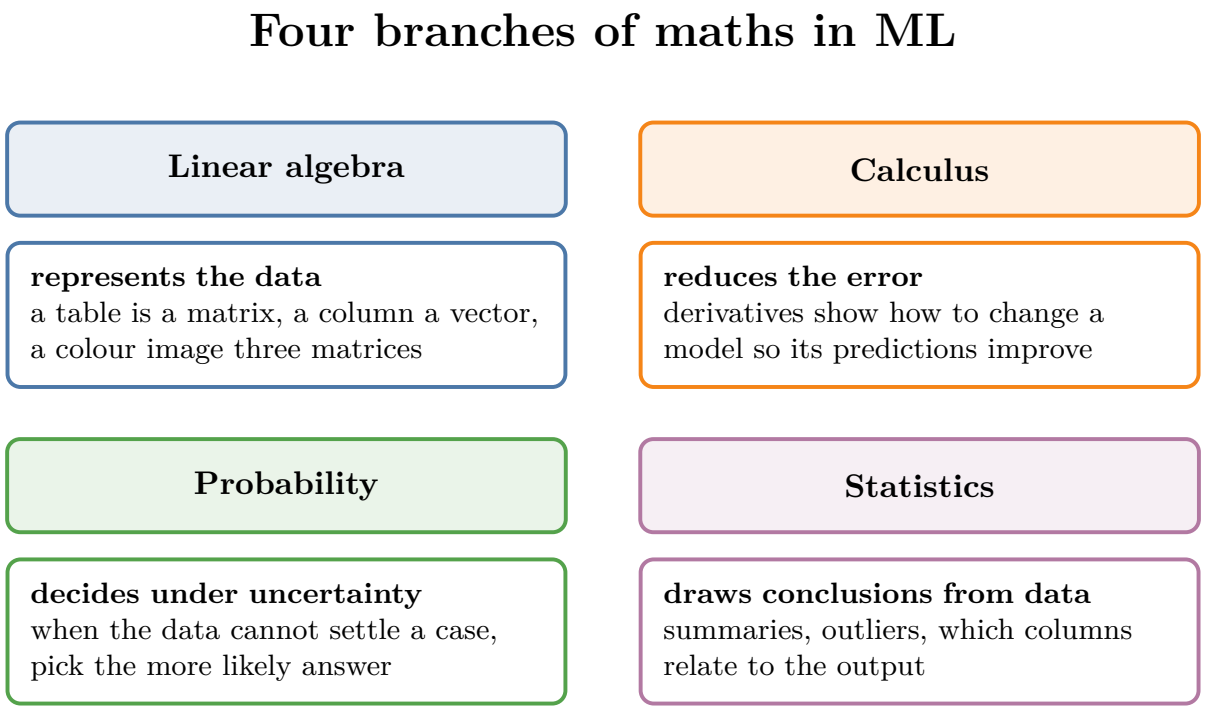


Figure 1: The four branches of mathematics used in ML, and the job each one does

Figure 1 shows the four branches and the job each one does. This Note gives a first feel for each job and points to the Notes where each branch is taught. The details come in those Notes.

2. Linear algebra: representing data

> **Key point:** Linear algebra gives us the containers for data (**vectors** (G-2079), matrices and tensors) and the operations that act on a whole container at once.

Linear algebra (G-1090) stores every kind of data (tables, text, images, video) as vectors (lists of numbers, such as $(2, 3)$), **matrices** (G-1180; tables of numbers) and **tensors** (G-1957; grids of numbers with any number of axes), and acts on a whole container in one step; why ML needs linear algebra explains why. How each kind of data becomes a tensor is shown in tensors from 0D to 5D.

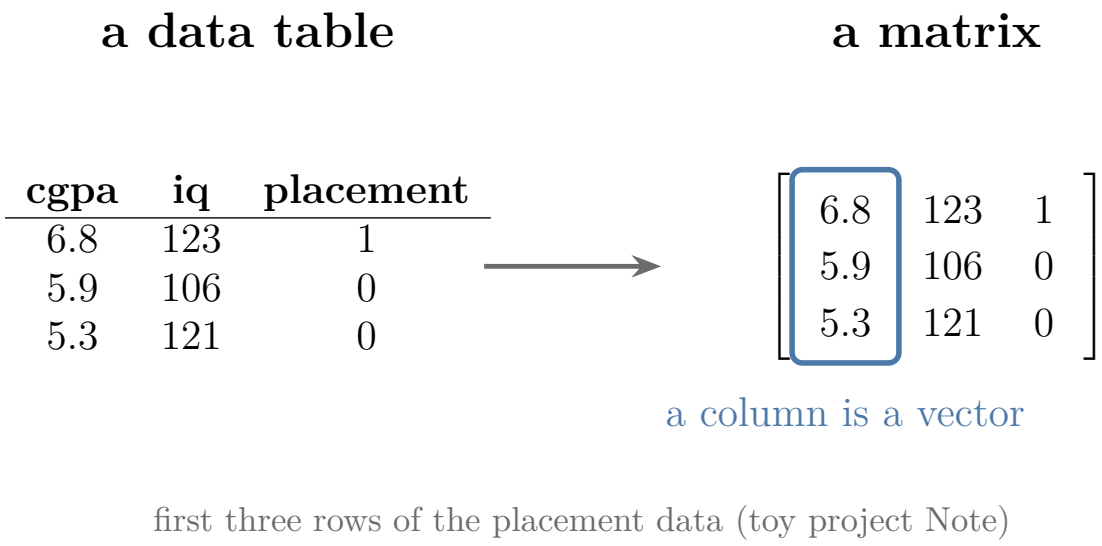


Figure 2: Three rows of the placement data as a table and as a matrix. Each column of the matrix is a vector

Figure 2 shows the first step of linear algebra in ML: the table we read becomes a matrix the computer can act on in one step.

The eight modules of the linear algebra roadmap list every linear algebra topic ML needs.

3. Calculus: reducing the error

> **Key point:** Predictions are never perfect; calculus tells us how to change a model so that its error shrinks.

Calculus (G-339) is the branch of mathematics about change: differentiation (slopes) and integration (areas). For beginners it is usually the hardest of the four.

An ML model predicts something, and its predictions are never exactly right. The gap between the prediction and the truth is the **error**. Training a model means changing it until the error is as small as we can get it, which is called **optimisation** (G-1400).

Most optimisation methods are built on calculus. The **derivative** (G-595) of the error tells us in which direction to change each part of the model so the error goes down. Gradient descent (G-862) does exactly this, one small step at a time.

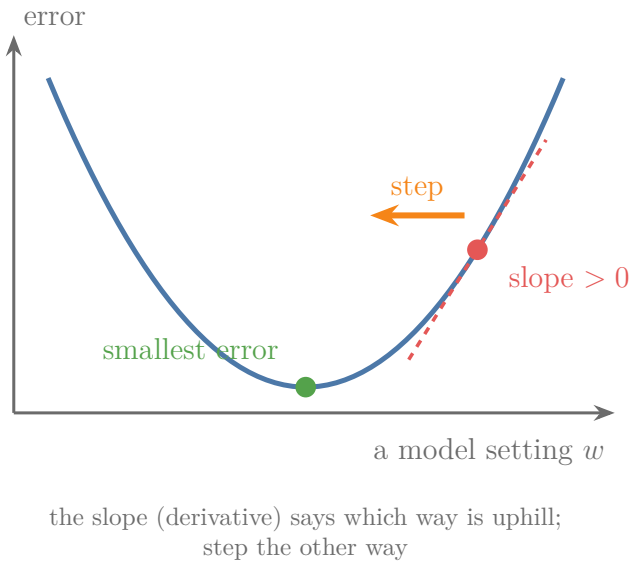


Figure 3: Error against one model setting w. At the red point the slope is positive, so a step to the left lowers the error

Figure 3 shows the idea: the sign of the slope says which way the error rises, and the model steps the other way, towards the green minimum. Figure 4 repeats the step until the slope is flat: each step is the slope times a small number (here 0.3), so the steps shrink as the model nears the minimum. Here w is one setting of the model, such as the slope of a line it fits, and the error curve is:

$$E(w) = (w - 2)^2 + 1$$

Starting at $w = 5$, the slope of the error curve is $2(w - 2)$, and each row below is one step (new w = old w minus 0.3× slope):

Step	w	Slope	Error
0	5	6	10
1	3.2	2.4	2.44
2	2.48	0.96	1.23
3	2.19	0.38	1.04
4	2.08	0.15	1.01
5	2.03	0.06	1.00
6	2.01	0.02	1.00
7	2.00	0.01	1.00

Step	w	Slope	Error

For example, step 1 moves w from 5 against the slope 6 (step size 0.3):

$$5 - 0.3 \times 6 = 3.2$$

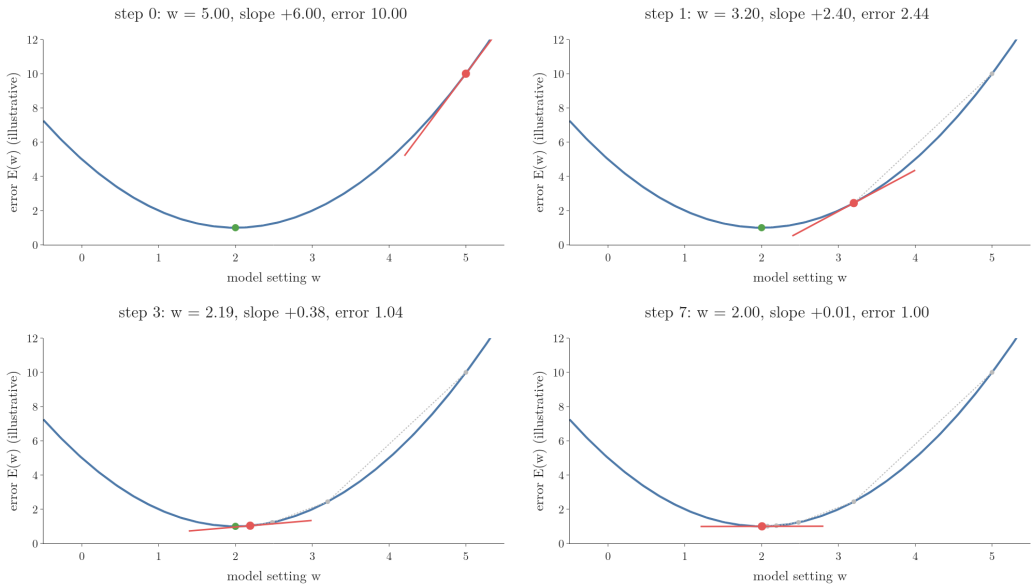


Figure 4: Gradient descent on an illustrative error curve $E(w) = (w - 2)^2 + 1$, from $w = 5$ with step size 0.3. The red line is the slope at the current point; the error falls from 10 to 1.00 in seven steps.

> **Extra:** A model with 100% accuracy is a warning sign, not a success. Real outputs contain a random part that no model can predict, called the **irreducible error** (G-1327; ISLR §2.1.1). So a perfect score often signals a mistake, such as test data leaking into training (**data leakage** (G-535), see scaling the inputs; Kaufman et al. 2012).

Calculus first appears in finding the minimum, where a derivative set to zero gives the best line. Derivatives themselves are taught step by step in from secant to tangent.

4. Probability: deciding in the grey zone

> **Key point:** When the data cannot settle a case for certain, probability tells us which answer is more likely, and we choose that one.

Often the data leaves a grey zone, where no answer is certain. Suppose we want to guess a person's gender from their salary alone. In the salary range 4 to 5 lakh rupees, there are both men and women, so salary cannot decide.

Probability (G-1574) still lets us decide. If 40% of the people in that range are men and 60% are women, then for a new person earning 4.5 lakh rupees the more likely answer is "woman", and that is what we predict. The chances inside one range add up to 100%.

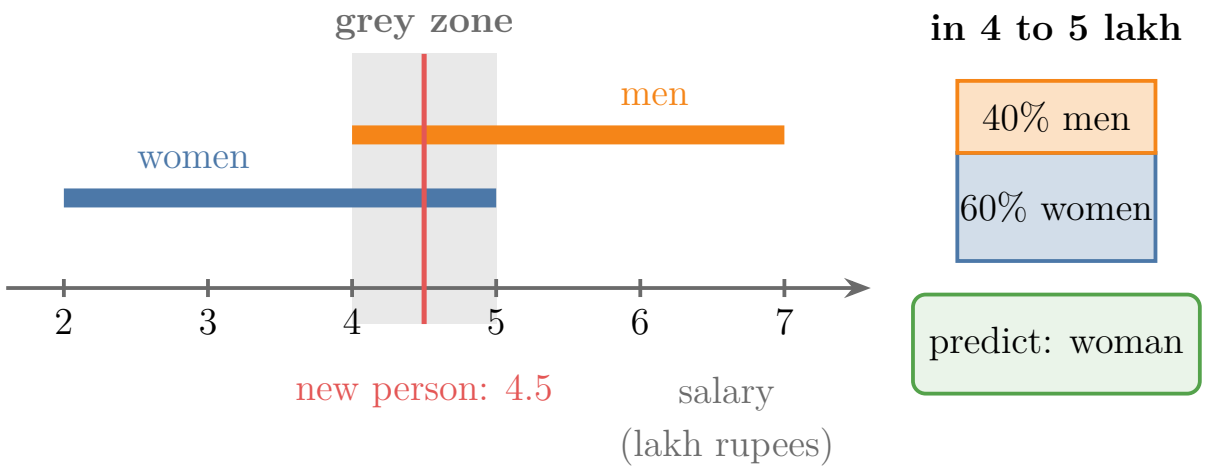


Figure 5: Salary cannot separate the two groups between 4 and 5 lakh rupees (the ranges outside that zone are drawn for illustration). With the shares 60% women and 40% men in that range, the prediction for a new person earning 4.5 lakh is "woman"

Figure 5 draws the grey zone: both groups overlap there, so we predict the class with the larger share.

Picking the most probable class is how probabilistic classifiers such as Naive Bayes (G-1297) work: compute the probability of every class, then pick the largest. The rules of probability are taught from the five basic terms of random experiments and events to joint, marginal and conditional probability, and **Bayes' theorem** (G-269) builds on them.

5. Statistics: drawing conclusions from data

> **Key point:** Statistics is used the most of the four: from the moment we receive data to the moment we use the result.

Statistics (G-1884) is the branch of mathematics for collecting and analysing data so that we can draw conclusions from it (see what statistics is). It appears at every stage of an ML project.

In real work, nobody hands us a dataset with a list of questions. We get the data and must find the meaningful questions ourselves, then answer them. Statistics supplies the tools:

- **Is there an outlier?** One value far from all the others can mislead a model (see how outliers spoil a model).
- **Is the data noisy?** Statistics helps us spot noisy data.

- **Which algorithm suits this data?** The shape of the data points to suitable algorithms.
- **Is this feature related to the target?** A **feature** (G-772) is an input variable (one column of the data table), and the **target** (G-1949) is the output we predict. If a feature is unrelated to the target, there is no reason to feed it to the algorithm. Measures such as **correlation** (G-490) answer this.

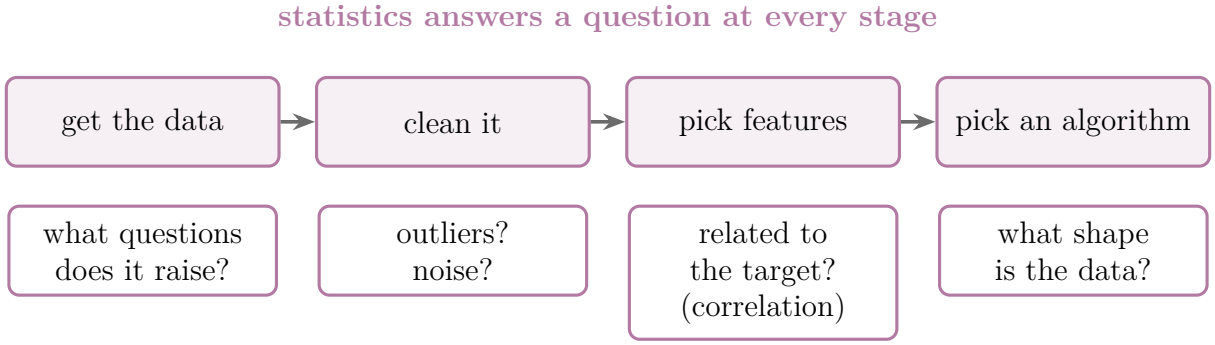


Figure 6: The questions statistics answers at each stage of an ML project

Figure 6 places the four questions along a project, from receiving the data to choosing the algorithm.

Data analysis, which ML depends on heavily, is built almost entirely on statistics (see what the data looks like in numbers). The four modules of the statistics roadmap map the whole subject.

6. Summary

Branch	Its job in ML	Example	Why it matters	Where it is taught
Linear algebra	Represent data and act on it	A table as a matrix	the computer acts on a whole table in one step	**Vectors** (G-2079), **tensors** (G-1957)
Calculus	Reduce the error (optimisation)	Gradient descent	the slope of the error says which way to change the model	**Gradient descent** (G-862)
Probability	Decide under uncertainty	Pick the more likely class	we can still decide when the data cannot settle a case	**Events** (G-717) onward
Statistics	Draw conclusions from data	Find outliers, related features	an outlier or an unrelated feature can mislead the model	Statistics roadmap

- ML is mathematics running on data; four branches carry most of it, so knowing them is how we understand why an algorithm works and fix it when it does not.
- Statistics is used the most, because it is needed at every stage, from receiving the data to choosing the algorithm; **linear algebra** (G-1090) does the heavy lifting of storing and transforming data, because every kind of data becomes vectors, matrices and tensors.

7. Sources

Built from

- CampusX, "Role of Mathematics in Machine Learning", YouTube, https://www.youtube.com/watch?v=

Other references

- James, G., Witten, D., Hastie, T. and Tibshirani, R. (2013). *An Introduction to Statistical Learning*. Springer. Section 2.1.1, reducible and irreducible error.
- Kaufman, S., Rosset, S. and Perlich, C. (2012). "Leakage in Data Mining: Formulation, Detection, and Avoidance". *ACM Transactions on Knowledge Discovery from Data* 6(4).

8. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Calculus (G-339)	The branch of mathematics about change: differentiation and integration.
Optimisation (G-1400)	Changing a model step by step until its error is as small as possible.
Linear algebra (G-1090)	The branch of mathematics that studies linear equations, vectors and matrices; ML uses it to store data (tables, text, images) as vectors and matrices and to work on a whole dataset in one step.
Vector (geometric view) (G-2079)	A point in a coordinate system, drawn as an arrow from the origin.
Matrix (G-1180)	A table of numbers in rows and columns (a 2D tensor); it can hold a dataset (one row per observation), and multiplying a vector by it transforms the vector.
Tensor (G-1957)	A container of numbers arranged along one or more axes.
Derivative (G-595)	The slope of a function at a point: how fast the output changes as the input changes there. It is zero at the lowest point of a smooth curve, which is how training finds a minimum.
Gradient descent (G-862)	Finding the lowest point of a function by repeated small steps downhill.
Noise (irreducible error) (G-1327)	Randomness in the data that no model can predict.
Data leakage (G-535)	Information from the test rows reaching the model during training, for example when a scaling or feature-selection step is fitted on all the data before the split; the test scores then come out higher than the model will really do on new data.
Probability (G-1574)	A number from 0 to 1 measuring how likely an event is; written $P(A)$.
Naive Bayes classifier (Naive Bayes) (G-1297)	A classifier that applies Bayes' theorem with the assumption that inputs are independent within each class.

Term	Meaning
Bayes' theorem (G-269)	The rule that reverses a conditional probability: from $P(B \mid A)$ it gives $P(A \mid B)$, so a belief about A can be updated after seeing B; $P(A \mid B) = P(B \mid A)P(A)/P(B)$.
Statistics (G-1884)	The branch of mathematics for collecting, analysing, interpreting and presenting data.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Target (G-1949)	The output column we predict, such as the class; also called the label.
Correlation (G-490)	A number from -1 to +1 that says how strongly two numerical features rise or fall together along a straight line: the covariance divided by both standard deviations, so the units drop out.
Event (G-717)	A set of outcomes, such as "the sum is at most 10".
Error (residual) (G-705)	The gap between an actual value and the model's prediction on one data point: actual minus predicted.